

FL Simulation Results

File: test_FoolsGold

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 3
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 100.0% (1.00)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 100.0% (1.00) of data for 15 rounds with 3 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8698
Clean + DP Protection Accuracy: 0.8696
Poisoned Accuracy: 0.4656
Poisoned + DP Protection Accuracy: 0.8280
Drop (Clean - Poisoned): 0.4041
Drop (Clean - Init): 0.0770
Drop (Clean DP - Init): 0.0768
Drop (Poisoned - Init): -0.3271
Drop (Poisoned DP - Init): 0.0352
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8771
Clean Recall: 0.8698
Clean F1 Score: 0.8692

Clean DP Precision: 0.8765
Clean DP Recall: 0.8696
Clean DP F1 Score: 0.8690

Poisoned Precision: 0.7063
Poisoned Recall: 0.4656
Poisoned F1 Score: 0.4837

DP Protection Precision: 0.8380
DP Protection Recall: 0.8280
DP Protection F1 Score: 0.8271

Summary

FL Simulation Complete
Task: aeafa723d-9900-4eca-90f9-50ffb43deddb
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8698
Clean DP Accuracy: 0.8696
Poisoned Accuracy: 0.4656
Data Poison Protection Accuracy: 0.8280
Drop (Clean - Poisoned): 0.4041
Drop (Clean - Init): 0.0770
Drop (Clean DP - Init): 0.0768
Drop (Poisoned - Init): -0.3271
Drop (Poisoned_DP - Init): 0.0352
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8771
Clean Recall: 0.8698
Clean F1 Score: 0.8692
Clean DP Precision: 0.8765
Clean DP Recall: 0.8696
Clean DP F1 Score: 0.8690
Poisoned Precision: 0.7063
Poisoned Recall: 0.4656
Poisoned F1 Score: 0.4837
DP Protection Precision: 0.8380
DP Protection Recall: 0.8280
DP Protection F1 Score: 0.8271